The Abstraction Frontier

A Pareto antichain of quotients trading task-sufficiency, dynamical closure, coding cost, and control regret

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Date: August 3, 2026 **Status:** two elementary theorems + one worked example (4-bit Boolean world). Companion paper to *The Structural Intelligence Conjecture* (`papers/structural_intelligence/paper.md`) and to *Sufficient Antecedents for Cross-Task Stability* (`papers/sufficient_antecedents/paper.md`); depends on Theorem 4 of the parent paper (cross-task stability via a shared Markov screen).

Abstract

Extended-program §5.3 of *The Structural Intelligence Conjecture* conjectures that the set of "reasonable" quotients `q : X → Z` for a task family `{Y_α}` is not a total order but a *Pareto frontier* trading four axes: task-sufficiency `I(Y_α; X | q(X))`, dynamical closure `I(Z_{t+1}; X_t | Z_t, A_t)`, coding cost `H_0(Z) = log|image(q)|`, and control regret. This paper turns that conjecture into two clean theorems:

- **Theorem AF-1 (Frontier is an antichain).** *The Pareto set of a candidate quotient lattice `Q` — quotients `q ∈ Q` such that no other `q' ∈ Q` weakly dominates `q` on all four axes and strictly on at least one — is an antichain in the product order. The Pareto set is monotone non-decreasing under lattice refinement: adding candidates to `Q` can only add (never remove) members from the frontier.*
- **Theorem AF-2 (Frontier reduces to CSS when tasks are decisive).** *If a common sufficient statistic `q*` exists for `{Y_α}` in the sense of Theorem 4 of the parent paper, then the Pareto frontier contains `q*`, and every Pareto member with zero task-sufficiency is at least as fine as `q*` on the lattice. If `q*` is unique and the dynamical-closure axis varies non-trivially with granularity, the zero-task-sufficiency portion of the frontier is a lattice segment rising from `q*` to the identity, ordered by coding cost. In the static case (dynamical closure and control regret constant across quotients), the zero-task-sufficiency portion collapses to `{q*}` alone: the identity is strictly dominated by `q*` on coding cost.*

The theorems are elementary: AF-1 is a definitional consequence of Pareto optimality on a product order; AF-2 is Theorem 4 restated on the Pareto-frontier side. Together they give a formal antidote to "the best quotient" thinking: two representations can be equally sufficient and equally cheap on some axes yet remain incomparable on the others, so neither dominates the other, and both are honestly on the frontier.

An exact instrument (`experiments/abstraction_frontier_pair`) exhibits the antichain on the 4-bit Boolean world of Instrument 4. The library of 23 quotients (concern-parameter lattice from that instrument) yields a **two-element frontier `{constant, joint(parity{0,1}, parity{2,3})}`**: the constant map (min-cost, worst-sufficiency endpoint) and the true `Z` (best-sufficiency, next-to-min-cost). The identity is strictly dominated by `Z`, and every single-bit or single-parity quotient is strictly dominated by the constant. This is the *static-case collapse* of AF-2, exhibited exactly and pre-registered as a gate.

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1. Setup

We inherit the master object of *The Structural Intelligence Conjecture* §1: a stochastic fibration `(q : X → Z, K : Z ⇝ X)` on a standard Borel `X`, together with a *task family* `{Y_α : α ∈ A}` on `X` and a *candidate quotient lattice* `Q` — a set of measurable maps `q : X → Z_q` (each with its own coarse space `Z_q`) that we consider as possible coarse-grainings.

For each `q ∈ Q` we define four scalar axes (all *lower = better*):

- **Task-sufficiency.** $TS(q) := \sup_\alpha I(Y_\alpha ; X \mid q(X))$, with the supremum taken over the task family. For deterministic $Y_\alpha = f_\alpha(X)$, sufficiency reduces to $\sup_\alpha H(Y_\alpha \mid q(X))$ under a fixed base distribution P on X. Zero iff q is a *common sufficient statistic* (CSS) for $\{Y_\alpha\}$.
- **Dynamical closure.** $DC(q) := I(Z_{t+1} ; X_t \mid Z_t, A_t)$, the residual information the coarse next-state $Z_{t+1} := q(X_{t+1})$ still carries about the current *fine* state X_t given the current coarse state $Z_t := q(X_t)$ and action A_t. Zero iff q is *self-contained* as a dynamical system (the coarse process is Markov on its own).
- **Coding cost.** $H_0(q) := \log_2 |\text{image}(q)|$, the description length of a single code from $\text{image}(q)$ in a uniform prefix-free code. This is the simplest granularity axis and lower-bounds any lossless code.
- **Control regret.** $CR(q) := V^*(X) - V^*(q(X))$, the loss in optimal value when a controller must condition its action only on $q(X)$ rather than on X. Zero iff a controller optimal on X factors through q.

Write $\text{axes}(q) := (TS(q), DC(q), H_0(q), CR(q)) \in \mathbb{R}_{\geq 0}^4$.

Standard Pareto. For $q, q' \in Q$, we say q' **weakly dominates** q iff $\text{axes}(q')_i \leq \text{axes}(q)_i$ for every axis i and $\text{axes}(q')_i < \text{axes}(q)_i$ for at least one axis. The **Pareto frontier** $F(Q)$ is the subset of Q that no other member of Q weakly dominates.

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2. Theorem AF-1: the frontier is an antichain

Theorem AF-1 (Frontier is an antichain). *$F(Q)$ is an antichain in the product order on $\mathbb{R}_{\geq 0}^4$: no two members of $F(Q)$ weakly dominate each other. Moreover, $F(Q)$ is monotone non-decreasing under lattice refinement: for $Q \subseteq Q'$, every member of $F(Q)$ that is not weakly dominated by some new candidate in $Q' \setminus Q$ remains in $F(Q')$, and $F(Q')$ may contain additional members from $Q' \setminus Q$.*

Proof.

(Antichain.) Suppose $q, q' \in F(Q)$ and q' weakly dominates q. Then by definition of the Pareto frontier, $q \notin F(Q)$, contradicting $q \in F(Q)$. So no two Pareto members weakly dominate each other, i.e. $F(Q)$ is an antichain.

(Monotonicity.) Let $Q \subseteq Q'$. Take $q \in F(Q)$. If no member of $Q' \setminus Q$ weakly dominates q, then no member of $Q' \supseteq Q$ does either (since q was Pareto in Q), so $q \in F(Q')$. Conversely, a member $q'' \in Q' \setminus Q$ is in $F(Q')$ iff no member of Q' weakly dominates it; there is no a-priori reason for this to fail, so $Q' \setminus Q$ can contribute additional Pareto members. Hence $F(Q')$ contains every $q \in F(Q)$ not dominated by a new candidate, plus possibly new members. □

Corollary (No total order). *For $|F(Q)| \geq 2$ there is no total order on $F(Q)$ induced by the product order on $\mathbb{R}_{\geq 0}^4$. Different frontier members correspond to genuinely different trade-offs and are incomparable.*

Remark (Trivial closure and identity endpoints). The constant map $q_\perp$ (image size 1) always has $H_0(q_\perp) = 0$; no other quotient can match it on coding cost, so $q_\perp$ is on the frontier whenever TS, DC, and CR at the constant are finite (which they always are for a bounded task family and finite V^*). The identity $q_\top$ (image X) always has $TS(q_\top) = 0$ on the deterministic-task case but has the maximum coding cost $H_0(q_\top) = \log_2 |X|$. Whether $q_\top$ is on the frontier depends on whether any other quotient matches $TS(q_\top) = 0$ at strictly smaller H_0 — precisely the content of Theorem AF-2.

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3. Theorem AF-2: the frontier and the CSS

Setup (Theorem AF-2). Assume the task family $\{Y_\alpha\}$ admits a common sufficient statistic $q^* : X \to Z^*$ in the sense of Theorem 4 of the parent paper: $Y_\alpha \perp\!\!\!\perp X \mid q^*(X)$ for every α, equivalently $TS(q^*) = 0$.

Theorem AF-2 (Frontier reduces to CSS when tasks are decisive). *Under this assumption:*

1. *$q^* \in F(Q)$.*
2. *Every $q \in F(Q)$ with $TS(q) = 0$ is at least as fine as q^* on the lattice of quotients of X (equivalently, q refines q^*).*

3. *If q^* is unique among CSS candidates in Q and if the dynamical-closure axis DC varies non-trivially with granularity (finer quotients strictly reduce DC), then the zero-task- sufficiency portion of $F(Q)$ is a lattice segment rising from q^* to the identity $q\_\top$, ordered by $H\_0(q)$.*
4. *In the **static case** — DC and CR constant across all $q \in Q$ — the zero-task-sufficiency portion of $F(Q)$ collapses to $\{q^*\}$ alone: any strictly finer sufficient quotient is dominated by q^* on coding cost.*

Proof.

(1) $q^* \in F(Q)$. Suppose some $q' \in Q$ weakly dominates q^*. Then $TS(q') \leq TS(q^*) = 0$, so $TS(q') = 0$; the domination is not on the TS axis. Since q^* has $H\_0(q^*) = \log\_2 |image(q^*)|$ and q' weakly dominates q^* on H_0, we have $|image(q')| \leq |image(q^*)|$, so q' is at least as coarse as q^*. But $TS(q') = 0$ means q' is also sufficient. A sufficient quotient no coarser than q^* (the coarsest CSS by Theorem 4's corollary) must equal q^* up to sufficiency-preserving refinement. Then q' is not *strictly* better than q^* on any axis, contradicting weak domination. So $q^* \in F(Q)$.

(2) Sufficient Pareto members refine q^*. Take $q \in F(Q)$ with $TS(q) = 0$. By Theorem 4's corollary applied to $Y\_\alpha \perp\!\!\!\perp X | q(X)$, $\sigma(q)$ is a sufficient sub-σ-algebra of $\sigma(X)$ containing $\sigma(Y\_\alpha)$ after the sufficiency reduction; combined with $\sigma(q^*)$ being the coarsest such, $\sigma(q^*) \subseteq \sigma(q)$, i.e. q refines q^*. So every sufficient Pareto member is at least as fine as q^*.

(3) The zero-TS segment. Under the two assumptions, the zero-TS portion consists of quotients refining q^*. Along that refinement chain $q^* \to q\_1 \to q\_2 \to \ldots \to q\_\top$, H_0 strictly increases (by definition of refinement adding new lattice cells) while TS stays zero. If DC strictly decreases along refinement, then no two elements of the chain dominate each other: coarser members have lower H_0 but higher DC, finer members lower DC but higher H_0. So every element of the chain is on the frontier, and the frontier's zero-TS portion is the chain itself.

(4) Static-case collapse. If DC and CR are constant across Q, then those two axes never separate quotients, and the effective Pareto reduces to (TS, H_0). Every strictly finer sufficient quotient $q > q^*$ has $TS = 0 = TS(q^*)$ and $H\_0(q) > H\_0(q^*)$; q^* weakly dominates q (equal on TS, DC, CR; strictly better on H_0). So $q \notin F(Q)$, and the zero-TS portion of $F(Q)$ collapses to $\{q^*\}$. □

Corollary (Two representations can be equally right yet incomparable). *Suppose $q\_1 \neq q\_2 \in F(Q)$ and both have $TS(q\_1) = TS(q\_2) = 0$. By AF-1, q_1 and q_2 do not weakly dominate each other, so they differ on at least one of DC, H_0, CR in opposite directions — q_1 is better on one, q_2 is better on another. Neither can be called "the right" quotient without further weight on the axes; the frontier itself encodes the honest trade-off.*

This is the formal antidote to *"the best quotient"* thinking. Two disentangled representations that agree on cross-task sufficiency can still legitimately disagree on how compact their code is or how self-contained their dynamics are, and both remain on the Pareto frontier as different balances of the same four axes.

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4. Worked example: 4-bit Boolean world

World. $X = \{0, 1\}^4$ (16 elements, uniform base distribution). Latent $Z(x) := (x\_0 \oplus x\_1, x\_2 \oplus x\_3)$, image size 4.

Task family (shared-through-Z from Instrument 4):

- $Y\_1 = parity\{0, 1\}$,
- $Y\_2 = parity\{2, 3\}$,
- $Y\_3 = parity\{0, 1, 2, 3\} = Y\_1 \oplus Y\_2$.

Each Y_α factors through Z, so Z is a CSS and $TS(Z) = 0$.

Quotient lattice. The concern-parameter lattice from Instrument 4 (`experiments/cross_task_sufficiency`), 23 candidates:

- `constant` (image 1, $H\_0 = 0$),
- 15 subset parities $parity\{S\}$ for $S \subseteq \{0, 1, 2, 3\}, S \neq \varnothing$ (image 2, $H\_0 = 1$),
- 3 joint pair-parities at image 4, $H\_0 = 2$ (`joint(parity{0,1}, parity{2,3}) = Z`, `joint(parity{0,2},`

`parity{1,3})`, `joint(parity{0,3}, parity{1,2}))`,

- 2 joint bit-reads at image 4, `H_0 = 2` (`joint(bit_0, bit_1)`, `joint(bit_2, bit_3)`),
- 1 joint triple-bit-read `joint(bit_0, bit_1, bit_2)` at image 8, `H_0 = 3`,
- `identity` (image 16, `H_0 = 4`).

Axes. `DC = CR = 0` for every `q` — the world is static and there is no controller, so these two axes are constants by convention. The effective Pareto reduces to `(TS, H_0)`.

Exact `TS`. Each `Y_α` is a balanced Boolean, so `H(Y_α) = 1` bit. Under the uniform base distribution:

- `TS(constant) = 1` (no information from `q`, every `Y_α`'s residual entropy is its own entropy 1).
- Any subset parity `parity{S}` reveals a single bit of information that is generically independent of the parity that defines `Y_α`; a direct computation gives `TS(parity{S}) = 1` for every `S` (one task is deterministic, the other two remain balanced).
- Only `Z` and any strictly-finer joint on `X` give `TS = 0`. Among the 23 candidates, exactly two do: `Z` itself (`H_0 = 2`) and the `identity` (`H_0 = 4`). Every other joint (non-`Z` pair-parity, joint bit-read, triple bit-read) has `TS = 1` (at least one `Y_α` is not determined by the joint).

Pareto set. With `DC = CR = 0`, standard Pareto on `(TS, H_0)`:

Quotient	`H_0`	`TS`	Dominator
`constant`	0	1	— (frontier)
every subset parity	1	1	`constant` (better `H_0`, equal `TS`)
non-`Z` pair-parities and bit-joints	2	1	`constant`
`Z = joint(parity{0,1}, parity{2,3})`	2	0	— (frontier)
triple bit-read	3	1	`constant`
`identity`	4	0	`Z` (better `H_0`, equal `TS`)

Frontier = `{constant, Z}`. A two-element antichain: the constant is incomparable with `Z` (better `H_0`, worse `TS`), and `Z` is incomparable with the constant (worse `H_0`, better `TS`).

This is the **static-case collapse** of Theorem AF-2 clause (4): even though `identity > Z` on the lattice with `TS(identity) = 0`, it is dominated by `Z` on coding cost, so the zero-`TS` portion of the frontier collapses to `{Z}`. In a dynamical setting where finer quotients strictly reduce `DC`, the identity (or intermediate refinements between `Z` and identity) would rejoin the frontier as alternative trade-offs.

Interpretation. The `constant` sits on the frontier because it is the cheapest representation to code (0 bits), but pays for that with maximal task-residual entropy. `Z` sits on the frontier because it is the *cheapest* sufficient representation (2 bits, strictly less than the 4-bit identity), and no coarser sufficient representation exists. Every other quotient in the 23-element lattice is honestly dominated — by the constant on the `TS = 1` plateau, by `Z` on the `TS = 0` plateau — and thus not on the frontier.

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5. Instrument: `experiments/abstraction_frontier_pair`

Exact witness of Theorems AF-1 and AF-2 on the setup above.

- Enumerate the 23-quotient library.
- For each `q`, compute `TS`, `H_0`, `DC := 0`, `CR := 0` exactly.
- Compute the Pareto frontier under standard weak-domination.
- Pre-registered gates (all six pass exactly):

- `af1_frontier_is_antichain`: no two Pareto members dominate.
- `af1_frontier_contains_true_Z`: `Z` is on the frontier.
- `af1_frontier_contains_constant`: the constant is on the frontier.
- `af1_identity_is_dominated_in_static_case`: `Z` weakly dominates the identity (equal `TS`, strictly better `H_0`).
- `af2_sufficient_frontier_is_true_Z_alone`: among Pareto members with `TS = 0`, the only one is `Z`.
- `af2_no_pareto_strictly_finer_than_true_Z`: no frontier member is strictly finer than `Z` on the lattice.

The instrument is deterministic (no random seeds, no Monte Carlo) and the frontier is `{constant, joint(parity{0,1}, parity{2,3})}` exactly.

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6. Relation to the SIC framework

Theorem AF-1 turns extended-program §5.3 from a *research direction* into a *theorem* about the shape of the reasonable-quotient set: an antichain, not a total order, monotone under lattice refinement. Theorem AF-2 links that shape back to Theorem 4 of the parent paper — the CSS `q*` is on the frontier and every zero-task-sufficiency frontier member refines it. Together with:

- Theorem CG-1 / CG-2 (concern as fiber geometry, Fisher metric and holonomy),
- Theorem CT-1 / CT-2 (compiler tomography, MDL identifiability and compiler ecology),
- Theorem SA-1 (antecedent taxonomy — four canonical `(U, P)` recipes populating Theorem 4),

the SIC extended program now has *six* explicit theorem-instrument pairs beyond the six of the parent paper. The remaining §5 constructs (theory atlas §5.5, causal semantics §5.7, representation-repair calculus §5.8, alignment as ensemble governance §5.9, autocatalytic artwork §5.10) remain open — each is a direction rather than a theorem.

The frontier framing sharpens the SIC honesty condition: representation choice is not a single quotient-level answer but an antichain of distinct trade-offs. Two disentangled representations can both be "right" and remain incomparable — the frontier itself is the ground truth.

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7. Limitations

- **Static-case collapse is generic.** On a truly static world, the zero-`TS` portion of the frontier reduces to `{q*}` alone; the "segment from `q*` to identity" statement of AF-2 clause (3) requires the dynamical-closure or control-regret axes to move non-trivially with granularity. In many applications those axes *do* move (state-abstraction MDPs are the canonical example), but the 4-bit example instrumented here does not exercise them. A Markov-decision-process instrument would exhibit clause (3) directly.
- **`DC` and `CR` are convention-set to 0.** In this instrument the two dynamical/controller axes are set to 0 for every quotient by convention; they carry no dominance information. A follow-up instrument on a Markov chain with induced coarse dynamics `(Z_t, A_t) → Z_{t+1}` would compute `DC` exactly and let those axes contribute to Pareto selection.
- **Task-sufficiency assumes finite tasks.** The definition `TS(q) := sup_α I(Y_α ; X | q(X))` is well-defined for a finite (or countable) task family; for a continuum family, the supremum may not be attained, and a metric on the task family is needed for a robust extension.
- **No Lean formalisation.** AF-1 is a definitional consequence of Pareto optimality on a product order, straightforward to formalise once product-order lemmas from mathlib are imported. AF-2 uses Theorem 4's Markov-screen argument (already formalised in `CommonSuffScreen.lean`); the reduction argument would slot on top. Both are future work.

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8. Reproduction

```
python3 experiments/abstraction_frontier_pair/experiment.py
```

Full development is in the parent paper's §5.3 and the notes file `notes/structural_intelligence_conjecture.md`.